Three-component decomposition of treatment response in aggregated N-of-1 trials: pharmacological, expectation-driven, and natural-history components

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Abstract

Background. Treatment response in aggregated N-of-1 clinical trials reflects three causally distinct mechanisms: pharmacological action of the active compound (BR), the patient's belief that the treatment will work (PB), and natural-history drift in the underlying condition (TV). Standard analysis models combine these into a single 'treatment effect' plus residual error. Whether this one-component approach biases inference, and whether a component decomposition repays its cost in degrees of freedom and power, is not unconditional: it depends jointly on the estimand, on the trial design, and on whether a candidate biomarker correlates with the non-pharmacological components. This paper makes that dependence precise and resolves it into a design-by-correlation-structure rule.

Methods. We formalize a three-component decomposition of the within-participant outcome trajectory, each component a modified Gompertz function with participant-specific random effects, and characterize the trial-design contrasts (open-label, blinded discontinuation, crossover, balanced-placebo) that supply identifiability. We derive the omitted-variable-bias identity for the one-component estimator, which expresses both the one-component treatment effect and the one-component biomarker slope as sums of component-specific terms. We then

evaluate analysis strategies in three Monte Carlo studies calibrated to the prazosin-PTSD reference parameters: Study A couples the biomarker to BR only ($c_{bm,PB} = 0$) and varies the unmodeled-component magnitudes (1,000 replicates per alternative cell, 5,000 per null, across $m_{PB} \in \{0, 1, 3, 6, 10\}$, $m_{TV} \in \{-1, 0, 1, 2\}$, $N \in \{35, 70, 100, 150\}$); Study B (100-replicate pilot) couples the biomarker to PB; and Study C (1,000 replicates) contrasts the standard hybrid design with a belief-decoupling balanced-placebo design across $c_{bm,PB} \in \{0, 0.1, 0.2, 0.3\}$.

Results. The identity shows that the one-component biomarker-treatment slope is displaced from the pharmacological slope by a term proportional to the biomarker's covariance with PB and TV, not by the magnitude of those components; three regimes follow. (i) *Biomarker orthogonal to PB and TV, hybrid design:* the one-component estimator is essentially unbiased across the whole grid (mean absolute bias 0.016 at $N = 35$, 0.006 at $N = 150$, within one Monte Carlo standard error (MCSE) of zero) and strictly more powerful than the phase-augmented decomposition (mean power 0.59 versus 0.35 at $N = 35$; 0.90 versus 0.62 at $N = 70$); decomposition removes no bias and only sacrifices power, so it is dominated. (ii) *Biomarker coupled to PB, standard hybrid design:* the one-component slope is biased in proportion to $c_{bm,PB}$ (one-component bias rising to +0.48, coverage falling to 0.86 at $N = 150$), but because the drug-state and belief contrasts are collinear under this design (analysis-stage correlation $\approx +0.6$), no decomposition recovers the pharmacological slope either. (iii) *Biomarker coupled to PB, balanced-placebo design:* adding covert-drug and open-placebo phases decouples the contrasts (≈ -0.66), and the belief-covariate decomposition then returns the unbiased pharmacological slope across the feasible contamination range (bias within one MCSE of zero, coverage 0.94–0.96) where the one-component estimator does not, at the cost of some power.

Conclusions. The benefit of decomposition is jointly determined by the biomarker's correlation structure and the trial design, not by either alone. When the candidate biomarker is orthogonal to PB and TV, a single-indicator analysis that exploits the design contrast is unbiased and most powerful, and a parametric component model is counterproductive. When the biomarker may correlate with PB or TV, the one-component slope is biased and decomposition is necessary, but the bias is recoverable only under a design that varies belief independently of drug exposure (balanced-placebo or open-hidden); on the standard hybrid design no analysis recovers it, so a biomarker-validation trial at risk of placebo-belief contamination must be fielded with the decoupling design. Component decomposition is likewise warranted whenever the estimand is the attribution of response to pharmacology versus belief versus natural history. A pragmatic specification for the cases that warrant it is `Sx ~ bm + t + Dbc + phase + Dbc:phase + bm:Dbc + bm:Dbc:phase` with random intercept and AR(1) residual correlation.

1 Introduction

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Decomposing treatment response into pharmacological, expectancy, and natural-history contributions is not a new idea. Pharmacometrics has long modeled disease progression and drug effect as separable longitudinal processes[1,2]; Kaptchuk and colleagues decomposed the placebo response itself into observation, ritual, and practitioner-relationship components in a three-arm IBS trial[3]; neuroimaging has directly dissociated pharmacological from expectancy-driven analgesia using objective neural signatures[4,5]. The partition has intuitive clinical appeal: a patient whose symptom improvement is pharmacological will benefit from long-term treatment even after novelty fades, whereas one whose improvement is primarily belief-driven may not sustain it as expectations normalize. The question this paper addresses is not whether the partition is biologically coherent – it is – but when it is necessary for valid inference.

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The decision criterion follows from an algebraic identity established in section 6: the one-component biomarker-treatment slope decomposes as $\beta_{bm}^{\text{lumped}} = \beta_{bm}^{BR} + w_{PB}\beta_{bm}^{PB} + w_{TV}\beta_{bm}^{TV}$, where the loadings w_{PB} and w_{TV} are determined by the trial design and the β_{bm}^{C} terms are component-specific slopes. Bias is zero if and only if $\beta_{bm}^{PB} = \beta_{bm}^{TV} = 0$, that is, if and only if the biomarker is uncorrelated with the non-pharmacological components. Whether that condition holds is determined by biology and domain knowledge, not by the magnitude of PB or TV in the study population. A population with a large placebo response but a biomarker orthogonal to it yields an unbiased one-component estimate; a population with a small placebo response but a biomarker that correlates with it yields a biased one.

COMT genotype exemplifies the first category. COMT modulates catechol-O-methyltransferase activity in prefrontal cortex, regulates dopaminergic signaling in expectancy circuits, and predicts placebo-response magnitude in IBS and chronic pain populations[6,7]. A trial using COMT as a predictive biomarker for prazosin response would face a one-component slope that conflates pharmacological prediction with expectancy moderation. CYP2D6 metabolizer phenotype exemplifies the second: the genotype determines hepatic drug metabolism and thereby plasma drug concentration, with no plausible pathway to expectancy or natural-history effects. ADRB2 Arg16Gly in asthma bronchodilator trials provides a further example: spirometric outcomes are objective and receptor binding affinity does not modulate belief. For both, the one-component pharmacological slope is already the target estimand.

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The paper proceeds as follows. Section 2 fixes compact notation and states the formal model. Section 3 develops each component with biological motivation, parametric form, and identifiability conditions. Section 4 maps the hybrid open-label-plus-blinded-discontinuation design's phase structure onto the decomposition's identifiability requirements. Section 5 develops the analysis-side methodology, including the phase-augmented linear mixed-effects model (LME) approximation. Section 6 works through the numerical illustration and derives the algebraic identity that anchors the conditional rule. Section 7 presents Study A (orthogonal arm) and the Study B pilot (coupling arm). Section 8 translates the conditional rule into a practical biomarker-validation decision criterion under the PATH framework[8]. No published N-of-1 trial has applied this evaluation framework; the paper's primary contribution is determining when decomposition changes the inferential answer rather than advocating for it as a universal default.

2 Notation and the formal model

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In this section we shall fix notation and state the formal model. The three components are defined precisely enough to make the identifiability and analysis-side arguments in later sections rigorous, but we have endeavored to keep the presentation compact. A fuller treatment, with worked numerical examples at each step, is available in the companion pedagogical document at `docs/24-component-decomposition-pedagogy.md`.

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For participant i at trial timepoint t:

$$Y_{it} \;=\; \mathrm{BL}_i \;-\; \left[\, BR_{it} + PB_{it} + TV_{it} \,\right] \;+\; \varepsilon_{it},$$

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Table 1: Principal psychometric and pharmacological terms used in the three-component decomposition.

Term	Meaning
BR	Biological response: pharmacological reduction in symptom score attributable to active drug.
PB	Placebo-belief response: reduction attributable to the patient's belief that they are receiving treat
TV	Time-variant component: reduction (or increase) attributable to the natural-history trajectory of
η(phase)	Expectancy multiplier for PB: 1 in open-label on-drug, ≈ 0.5 in blinded discontinuation, between
BL_i	Participant i's baseline (pre-treatment) symptom score.
D_{it}	Continuous-decay drug indicator on the analysis side: 1 on-drug, $\exp(-\lambda t_{sd,it})$ off-drug.
$t_{sd,it}$	Time since drug discontinuation (off-drug only).
λ	Carryover decay rate, with half-life $t_{1/2} = \ln 2/\lambda$.

Each component is a participant-specific function of its own time-on-state variable. BR_{it} depends on time-on-drug, the cumulative exposure to active medication since drug initiation. PB_{it} depends on time-on-belief, modulated by a phase-specific expectancy factor η(phase) that captures the patient's confidence that they are receiving active treatment. TV_{it} depends on time-since-trial-entry, calendar time elapsed regardless of treatment status.

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Each component follows a modified Gompertz curve:

$$C(t) \;=\; m_C \exp(-d_C \exp(-r_C t)) \;-\; m_C \exp(-d_C),$$

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ε_{it} is a within-participant residual term with AR(1) serial-correlation structure $\mathrm{Cov}(\varepsilon_{it}, \varepsilon_{i,t+h}) = \sigma_\varepsilon^2 \rho^{|h|}$. The three components are permitted to be jointly correlated through within-time and across-time cross-component covariances c_{cf1t} and c_{cfct} respectively, so that participant-level heterogeneity in any one component can co-vary with heterogeneity in the others.

The principal psychometric and pharmacological terms used below are summarized in Table 1.

3 The three components

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In this section we shall present the three components in turn, each with its biological motivation, the modified Gompertz parameterization, and the design contrasts that identify it. The treatment is organized so that the reader can see, by the end of this section, that the formal model is a sum of three causally interpretable trajectories rather than a mathematical convenience.

3.1 Biological response (BR)

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BR is the physiological reduction in symptom score attributable to the active medication's chemical action on the body. It is independent of belief, expectancy, and natural history: a participant who somehow received active drug without knowing it would still develop a measurable BR, and a participant who took an inert sugar pill while believing it was active drug would have $BR = 0$ no matter how strong their expectation. Methodologically, BR is the quantity that drug regulators and prescribers want: approval decisions, dose selection, and biomarker-guided patient stratification are all framed around BR.

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The qualitative shape of the BR trajectory follows from pharmacokinetic and pharmacodynamic considerations. Drug concentration in plasma reaches steady state within a small multiple of the elimination half-life; brain concentration follows plasma with a short delay; the downstream pharmacodynamic response (reduction in symptom-relevant neural activity, in the case of prazosin's effect on noradrenergic tone during REM sleep) typically requires several days to weeks of sustained exposure to integrate fully[9]. The result is a saturating curve: slow rise in the first week, steepest growth in the second to third week, plateau as receptor adaptation completes, and a decay back toward zero on discontinuation governed by the carryover half-life $t_{1/2} = \ln 2/\lambda$ rather than an instantaneous drop. The rate of this off-drug decay is the object of the carryover machinery used throughout this project; only in the limit of a short half-life relative to the measurement spacing does it approximate the sharp step assumed in the worked example of section 6.

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The modified Gompertz form $BR(t) = m_{BR}\exp(-d_{BR}\exp(-r_{BR}t)) - m_{BR}\exp(-d_{BR})$ captures this profile compactly. The asymptote $m_{BR}(1-e^{-d_{BR}})$ is the biological ceiling, the largest reduction the drug can produce at infinite exposure; the scale parameter m_{BR} equals that ceiling up to the factor $(1-e^{-d_{BR}})$, which for the calibrated $d_{BR} \approx 5$ exceeds 0.99. The onset rate r_{BR} controls how rapidly the participant approaches the ceiling: high r_{BR} means a fast onset (effect plateaus in one to two weeks), low r_{BR} a slow climb (six to eight weeks). The displacement d_{BR} shapes the climb: high d_{BR} gives an early-onset curve that rises sharply at first, low d_{BR} a more gradual climb. For PTSD prazosin, calibrated values from the[10] reference dataset are $m_{BR} \approx 11$ (eleven points of nightmare-score reduction at saturation), $r_{BR} \approx 0.42$ per week (half-maximum at roughly four to five weeks of sustained exposure), and $d_{BR} \approx 5$ (moderately steep early climb).

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Two alternative families that do not capture the asymmetric saturation profile are worth noting for contrast. A linear function $BR = \beta t$ climbs without bound and is biologically implausible past a few weeks. An exponential decay $BR = m_{BR}(1-\exp(-rt))$ is acceptable for fast-onset drugs but lacks the slow-start phase that characterizes most chronic-condition responses, where receptor adaptation, feedback regulation, and gene-expression changes produce a measurable lag. The Gompertz interpolates between exponential decay (high d limit) and linear-in-t growth (low d limit), and its three-parameter form is flexible enough to fit a wide range of pharmacodynamic profiles without overfitting. The Gompertz is a deliberate choice over the two parametric forms most familiar from pharmacokinetic-pharmacodynamic modeling. The sigmoid $E_{\max}$ (Hill) model[11] describes the dependence of effect on concentration (a dose-response curve), whereas $BR(t)$ here is a time-course of accumulating effect at fixed dosing; and the indirect-response (turnover) models that do describe effect time-courses are specified through differential equations on a hidden response variable rather than in closed form. The modified Gompertz retains the slow-start, saturating shape those models produce while remaining a closed-form, three-parameter mean function that is tractable inside a mixed-effects likelihood. Doc 25 of the project documentation provides the full historical and biological context for the family choice and quantifies the sensitivity of pmsimstats inference to alternative families; that work is reported separately as `07-gompertz-evaluation`. The headline result of that evaluation

is reassuring for the present framework: in a 1{,}000-replicate-per-cell simulation that generated data under the logistic, hyperbolic-tangent, and piecewise-linear breakpoint families and analyzed them under the standard linear-mixed model, the trajectory family was essentially irrelevant to the power of the biomarker-treatment interaction test (the family-to-family spread fell within two MCSEs of sampling noise) and to Type I error (0.025 to 0.036 across null cells, with no family-specific inflation). The conclusions drawn here are therefore not artifacts of the Gompertz form, though the regimes the family cannot represent (cyclical or biphasic trajectories) remain genuine exclusions.

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The BR component is identified by within-participant on-drug versus off-drug contrasts, formalized in section 4 below. The most direct contrast is the blinded-discontinuation phase: when a participant is silently switched to placebo while still believing they are on active drug, BR decays toward zero as the drug clears while PB remains roughly intact, and the difference between on-drug and off-drug response in that window is a relatively clean estimate of BR that does not require the analyst to model PB or TV explicitly. The decay is not instantaneous: with the carryover half-life $t_{1/2} = \ln 2/\lambda$ of the analysis-side decay term, residual BR persists into the early off-drug observations, so the contrast is clean only once the discontinuation window is long relative to $t_{1/2}$, and the drug state is best carried by the continuous-decay indicator D_{it} of Table 1 rather than a hard on/off indicator.

3.2 Placebo-belief response (PB)

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PB is the reduction in symptom score that arises from the patient's belief that they are receiving active treatment, independent of what they are actually receiving. That is, it is the placebo response in its strict sense: the part of the improvement that would persist if the active drug were silently replaced with an inert pill, provided the patient continued to believe they were on the active drug.

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It is worth being precise about what PB is and is not. PB is not 'fake' improvement; it is a phenomenon clinicians knowingly exploit, a national survey of US internists and rheumatologists finding roughly half report regular use of placebo treatments[12]. As the introduction sets out, the placebo response is mediated by well-characterized neurobiological pathways[13–17]: belief-driven release of endogenous opioids and dopamine, anticipatory hypothalamic-pituitary-adrenal regulation, and top-down attentional shifts, dissociated from pharmacology by the[4] neurologic pain signature[5] and shown to have genetic correlates[6,7]. What makes these mechanisms 'placebo' rather than 'drug' is that they are triggered by belief rather than by pharmacology.

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For trial design, the relevant point is that PB is separately identifiable from BR if and only if the trial includes a phase in which the patient's belief about treatment differs from the actual treatment they are receiving. The blinded-discontinuation phase is the canonical such phase: the patient is told that, at some unannounced point during the window, they may be silently switched to placebo, and they cannot reliably tell whether they are still receiving active drug. Belief in treatment correspondingly drops to a partial level (multiplier roughly 0.5) because the patient's expectation is hedged.

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PB also follows a Gompertz curve, scaled by the expectancy factor:

$$PB(t \mid \text{phase}) \;=\; \eta(\text{phase}) \cdot m_{PB}\, \exp(-d_{PB} \exp(-r_{PB} t)) + (\text{offset to start at zero}).$$

Typical expectancy values are $\eta = 1.0$ in open-label on-drug, $\eta \approx 0.5$ in blinded discontinuation, and η between 0 and 0.5 in open-label placebo (the residual expectation a patient retains when told the previous active phase was beneficial). The quantity η operationalises response expectancy in the sense of[18], the patient's anticipation of benefit that the expectancy literature treats as a determinant of experienced symptom change. The choice of $\eta = 0.5$ for the blinded phase is a modeling assumption rather than a calibrated quantity: to our knowledge the residual expectancy a patient carries when uncertain about allocation has not been measured directly as a fraction of the open-label level, and we adopt 0.5 as a deliberately central placeholder. The available indirect evidence is nonetheless that

the residual response is substantial. Open-label placebo trials[19–21] demonstrate that a clinically meaningful response persists even when patients are explicitly told they are receiving an inert intervention, which bounds the residual belief-driven response away from zero. Because the value of η is not empirically pinned, sensitivity analyses varying it across its plausible range are mandatory before any conclusion is allowed to depend on this single number.

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The decomposition framework as proposed here has a structural analogue in the placebo-research literature.[3] decomposed the placebo response itself into observation, ritual, and patient-practitioner relationship components in a randomized trial of acupuncture-style placebo for irritable bowel syndrome, demonstrating empirically that the placebo response is not a single 'expectancy' lump but a sum of identifiable parts. The present BR-PB-TV framework is analogous in spirit but operates at the level of the full treatment response (drug, placebo, natural history) rather than within the placebo response alone.

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A subtle property of PB is that its variance also scales with η, not just its mean. When η is high, there is substantial variability across patients in how strongly the placebo response is expressed; high-expectancy patients show large PB, low-expectancy patients show essentially none. When η is low (blinded), the placebo response is uniformly attenuated and the patient-to-patient variance shrinks accordingly. The model therefore parameterizes $\mathrm{SD}(PB \mid \text{phase}) = \eta(\text{phase}) \cdot \sigma_{PB}^{\text{baseline}}$. Ignoring this heteroscedastic structure deflates standard errors in the open-label phase and inflates them in the blinded phase, with downstream consequences for confidence-interval coverage and for the calibration of any biomarker test built on top of the analysis.

3.2.1 Additivity, subadditivity, and the BR-PB interaction

[Extended treatment in supplement S2.]

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The additive decomposition of section 2 treats the three components as independent contributors. Subadditivity – where the combined $BR+PB$ effect falls below the sum of the components measured separately – is documented in some clinical contexts[3,14,22] and can arise from shared neural substrate, ceiling effects, or belief-dynamics adaptation. A generalized model with $\gamma \cdot BR_{it} \cdot PB_{it}$ is identifiable from the same phase contrasts. Apparent subadditivity can also arise from latent-class structure with class-correlated components rather than a direct BR-PB interaction; disentanglement requires a class-aware analysis or the phase contrasts provided here. The phase-augmented analysis of section 5 absorbs subadditivity partly as a side benefit.

3.3 Time-variant natural-history component (TV)

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TV is the reduction in symptom score that would have happened even if the patient had received no treatment at all. It is the participant's natural-history trajectory, integrated over the duration of the trial. For some participants TV is positive (symptoms naturally improve over the trial because of life events, therapy, time-in-condition, or regression to the mean); for others TV is negative (symptoms naturally worsen because of trauma anniversaries, accumulating stress, or progressive underlying disease).

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Like PB, TV is not noise. It is a structured participant-specific signal that the model can in principle estimate from data, given sufficient repeated measurements and a trial design that includes a long enough off-drug baseline or follow-up window. What distinguishes TV from PB is its origin: TV is independent of treatment status and of belief about treatment, whereas PB is modulated by both.

3.3.1 Natural history: existence, testability, and functional form

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The shape of TV depends on disease class, trial timescale, and enrollment selection. Acute conditions trend TV-positive (intrinsic recovery); chronic stable conditions trend near $TV = 0$ on the population mean but with substantial participant-level variance; progressive conditions trend TV-negative. Selection-driven enrollment adds a positive TV component regardless of class, because patients enrol at moments of peak severity and regression to the mean then contributes apparent improvement in early trial weeks[23,24]. The zero-TV assumption is testable: a model including TV returns $\hat{m}_{TV}$ with a variance estimate; folding TV into ε is worse, because the participant-level natural-history trajectory is structured and correlated with the on/off-drug schedule, producing autocorrelated within-participant residuals and heteroscedastic between-participant residuals that bias the standard errors on BR and PB. A simple `week` covariate is insufficient: it enforces a linear, population-mean trajectory and, without the design contrasts of section 4, absorbs both BR and TV into a single coefficient that identifies neither.

4 Identifiability and trial-design contrasts

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The mathematical decomposition above is meaningless unless the data contain enough information to pin down each component separately. Trial design is what supplies that information. In this section we shall characterize the hybrid open-label-plus-blinded-discontinuation design, used as the worked example throughout this paper, and show how each of its three phases contributes a different combination of components to the observed trajectory.

Phase	Drug?	Belief?	Components present
Open-label on drug	Yes	Full	$BR + PB(\eta = 1) + TV$
Blinded discontinuation	Hidden	Hedged	$BR \cdot \mathbf{1}_{\text{on drug}} + PB(\eta = 0.5) + TV$
Open-label crossover	Either	Knows	$BR \cdot \mathbf{1}_{\text{on drug}} + PB(\eta = 1) + TV$

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Three phase-by-allocation contrasts then identify the three components. The open-label versus blinded contrast (within drug) yields $PB(\eta = 1) - PB(\eta = 0.5) \approx 0.5 \cdot PB$, which isolates the belief component, but only conditional on a known blinded-phase expectancy η: from a two-phase design just the product $\eta \cdot m_{PB}$ is identified, and separating η from m_{PB} requires the open-label placebo phase. The blinded on-drug versus blinded placebo contrast yields BR once drug carryover has elapsed, since PB and TV match across allocation arms within the blinded phase. All phases off drug yield TV plus any residual off-drug PB, with no active BR contribution beyond carryover and PB at the off-drug expectation level. The $\mathbf{1}_{\text{on drug}}$ indicator in the phase table above is shorthand for the continuous-decay drug state D_{it} defined in Table 1.

This conditioning on η is the framework's principal identification limitation and deserves emphasis rather than a parenthetical. With a two-phase (open-label and blinded) design the data identify only the product $\eta \cdot m_{PB}$, so PB is recovered up to the assumed blinded-phase expectancy; the working value $\eta \approx 0.5$ is a modeling assumption, not an estimate, and absent a belief or expectancy measurement it is not falsifiable from the outcome data alone. The identified set for m_{PB} is accordingly the interval $\{\widehat{\eta m_{PB}}/\eta : \eta \in (0, 1]\}$, which collapses to a point only when η is pinned down. Two design features close the gap: an open-label placebo phase, which supplies a second equation separating η from m_{PB}, and direct measurement of belief, as in the open-hidden and balanced-placebo designs[25,26] that manipulate or record the patient's treatment expectation. We therefore recommend that any trial relying on the decomposition for a PB-versus-BR attribution either include such a phase or collect an expectancy measure, and that analyses report sensitivity of the attribution to η across its plausible range.

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A trial that lacks any of these phases loses the corresponding identifiability. A pure open-label design (no blinded discontinuation) cannot separate BR from PB, because both are present at full strength throughout. A pure blinded design (no open-label phase) cannot estimate $\eta = 1$ PB, because no phase has belief at full strength. The decomposition machinery is useful only to the extent that the design supplies identifiability, and trial designers should evaluate any proposed design by checking that each component has at least one phase or contrast in which it is uniquely present.

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Identifiability in the design-contrast sense above is necessary but not sufficient for stable estimation at finite sample sizes. The three Gompertz trajectories share an onset shape and overlap in calendar time, so their parameters can be collinear even when the contrasts that identify them in principle are present. Two conditions make the joint estimation credible: the design must supply at least one phase in which each component varies while the others are held fixed (the blinded-placebo contrast for BR, the open-label-versus-blinded contrast for PB, and an off-drug window for TV), and that off-drug window must be long enough relative to the carryover half-life for TV to separate from a still-decaying BR. Where these conditions are weak the components are jointly identified only up to a near-collinear direction, and the symptom of that is the rank-deficiency the pilot of section 7 encountered when the full phase-augmented formula was fitted at $N = 35$. We therefore do not assert that the three components are always recoverable. The identifiability diagnostics of section 7 (convergence rates of the component-matched fit, variance-inflation factors on the component-specific Gompertz parameters, and the sensitivity of the BR estimate to the assumed η(phase) multipliers) are the means by which the recoverable region of the design-and-sample-size space is to be mapped, and the framework's participant-level claims should be read as conditional on falling inside that region.

5 Analysis-side methodology

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In this section we shall take up the practical question of what analysis model to use. A reasonable reaction to the formal model in section 2 is the following: if we are simulating data with three components plus noise, why is the analysis model written as `Sx ~ bm + t + Dbc + bm:Dbc` with a single random intercept and AR(1) residual correlation? Should we not include three parametric Gompertz components in the analysis, mirroring the DGP, so that the analyst sees the same structure that generated the data? The question is reasonable. Four constraints make a component-matched analysis impractical for most aggregated N-of-1 trial sizes, and a modest extension (a phase-by-treatment indicator) captures the practically important part of the gap at minimal cost.

5.1 Four reasons not to match the DGP

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The first reason is identifiability. The trial design must supply the contrasts that identify each component, and many designs supply only part of it. The decomposition is identified by phase contrasts (open-label versus blinded for PB, on-drug versus blinded-placebo for BR, off-drug timepoints for TV). A pure open-label trial has none of these contrasts; in such a trial BR and PB both rise from zero on treatment initiation and are statistically indistinguishable from each other, regardless of how the analyst writes the model. Forcing an analysis to fit three components when the design supplies contrasts for only one or two of them does not recover the missing components; it produces unidentified or weakly identified parameter estimates with implausibly tight standard errors that misrepresent the analyst's actual inferential precision.

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The second reason is degrees of freedom. Each component term costs degrees of freedom that come out of the moderation estimand. The biomarker-moderation question (the core scientific aim of the predictive-biomarker validation program) is asked through the `bm:Dbc` term. A component-matched analysis adds a Gompertz BR (three parameters), a Gompertz PB (three parameters plus a phase-specific expectancy multiplier), and a Gompertz TV (three parameters), each with participant-specific random effects on the maxima. That is on the order of a dozen new parameters, half of them on a non-linear scale. At trial sample sizes of $N \in [30, 150]$, the variance of $\widehat{\beta}_{bm:D}$ is dominated by participant and timepoint counts, and adding parametric structure to absorb residual confounding usually inflates the moderation estimand's variance by more than it reduces bias.

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The third reason is convergence fragility. Component-matched analyses are non-linear in the parameters and their convergence is problematic in practice. Fitting Gompertz BR, PB, and TV simultaneously requires a non-linear mixed model (`nlme::nlme` with self-starting Gompertz forms), a Bayesian hierarchical fit, or a structural-equation approximation. Each is more sensitive to starting values, more prone to local optima, and more likely to fail to converge on a single trial dataset than the linear-mixed analogue. At trial-relevant N, the convergence rate of the component-matched fit is typically below 90%, and the non-converging cells contaminate any aggregated analysis.

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The fourth reason is that the moderation estimand is sufficiently robust to the BR-versus-PB split for the most common scientific question. For predictive-biomarker validation, the regulator wants to know whether the biomarker predicts treatment response under the trial conditions, not a clean pharmacological-versus-placebo decomposition of that prediction. The simple `bm:Dbc` interaction is the right marginal estimand for the validation question whether the moderation is biological or expectation-mediated.

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It is worth stating the inferential target of the analysis in the vocabulary of the estimands framework[27], because the biomarker-validation question is easy to pose ambiguously. The primary estimand is the population-level biomarker-treatment interaction, the $bm{:}D_{bc}$ coefficient: its population is the trial-eligible patient population, its treatment contrast is on-drug versus off-drug exposure summarized through the continuous indicator D_{bc}, its endpoint is the symptom trajectory, its population-level summary is the interaction coefficient, and its intercurrent events are handled by a treatment-policy strategy for dropout and by the modeled exponential decay for residual carryover. The pharmacological-component target β_{bm}^{BR} is a more ambitious estimand: it is the biomarker-by-BR interaction that would obtain with the belief-driven channel held at its off-drug level, identified here through the phase contrasts of section 4 rather than by assumption. We flag explicitly that recovering a participant-level interaction of this kind inherits the difficulty Senn has emphasized for individual treatment effects[28,29]: a genuine participant-by-component signal is separable from participant-by-period and within-person random variation only when the design supplies repeated phase contrasts, and the framework should be read as estimating these quantities under that condition rather than as dissolving the well-known problem of identifying individual effects from finite within-person data.

5.2 A small extension that usually pays off: phase indicators

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The smallest useful step toward component-aware analysis, without paying the costs above, is to add a phase indicator and its interaction with the drug indicator. Schematically,

```
Sx ~ bm + t + Dbc + phase + Dbc:phase
       + bm:Dbc + bm:Dbc:phase
       + (1 | ptID),  correlation = corCAR1(...)
```

Here `phase` is a categorical covariate (open-label, blinded, post-crossover) that absorbs the systematic differences in PB expectancy across phases. `Dbc:phase` tests whether the apparent drug effect shrinks under blinding, a signature that the open-label drug effect was partly PB-mediated. `bm:Dbc:phase` tests whether the moderation itself shrinks under blinding, a signature that the biomarker-treatment interaction was partly biomarker-by-PB rather than biomarker-by-BR.

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This extension adds three to four parameters at most, well within budget at any realistic N. It does not separately estimate BR and PB as components, but it does identify the change in apparent drug effect that occurs when belief is hedged. For most analyses, this is the practically relevant question, and it can be answered without committing to the Gompertz parametric form. Where both the one-component and the phase-augmented analyses are reported, the additional `Dbc:phase` and `bm:Dbc:phase` terms introduce a small family of belief-attenuation tests, and the attribution inference they support should be treated as a secondary analysis with its multiplicity acknowledged rather than as a second bite at the primary interaction test.

5.3 When to fit the full decomposition

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A component-matched analysis merits the effort when sample size is in the several hundreds (a phase-3 trial fielded as aggregated N-of-1, or a meta-analysis pooling multiple smaller trials); the trial design provides timepoints in every phase combination, with enough density per cell to constrain the Gompertz curves; the scientific question requires the BR-PB split (a pharmacology paper claiming a specific mechanism, where ruling out a placebo contribution is the central point); and

a Bayesian hierarchical fit with informative priors on the Gompertz parameters is feasible. In that regime, `lcmm`, `nlme::nlme` with self-starting Gompertz forms, or a `brms` hierarchical formulation can return participant-specific BR, PB, and TV component estimates that mirror the DGP. Outside that regime, the linear-mixed analysis with phase indicators is the better tool.

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The asymmetry between DGP and analysis is not a flaw of the framework; it is a deliberate consequence of the asymmetry between simulation (where the analyst controls the DGP and can specify any structure, however elaborate) and inference (where the analyst must fit a model to a finite dataset and pay the identifiability cost of every additional parameter). Simulation studies use the rich DGP to characterize how a simpler analysis behaves under realistic data generation; they do not require the analysis to match the DGP in order to produce useful inference.

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A word on the carryover parameterization is in order, since it is the one place where the analysis model makes a strong functional-form commitment. The continuous drug indicator D_{bc} decays as $\exp(-\lambda t_{sd})$ off drug, a one-parameter exponential governed by the carryover half-life. This is deliberately parsimonious and is best understood as a constrained special case of the more flexible carryover models now available for repeated-measures and N-of-1 data. Liao et al.[30] model carryover through a Bayesian distributed-lag structure with autocorrelated errors, relaxing the single-rate exponential assumption at the cost of additional parameters; Daza[31] formalizes an average-period treatment-effect estimand built to accommodate autocorrelation, trend, and slow carryover jointly; and G{"a}rtner et al.[32] benchmark linear models against G-estimation and Bayesian-network alternatives specifically under carryover. We adopt the exponential form because it ties directly to a pharmacokinetic half-life and conserves degrees of freedom for the moderation estimand at trial-relevant sample sizes, but the choice is an assumption rather than a finding. The simulation program of section 7 should accordingly include a distributed-lag analysis arm, so that the cost of the exponential-decay restriction is quantified against the richer alternative rather than asserted.

6 Worked example: information loss without decomposition

6.1 What the one-component analysis estimates

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Before the numerical illustration it is worth establishing algebraically what the one-component analysis actually estimates, because the qualitative failure modes of section 1 are consequences of an identity rather than empirical tendencies that require simulation to demonstrate. Write the within-participant symptom reduction as $R_{it} = \mathrm{BL}_i - Y_{it} = BR_{it} + PB_{it} + TV_{it} - \varepsilon_{it}$. The one-component analysis fits a single drug indicator D_{it},

$$R_{it} \;=\; \alpha + \beta_D\, D_{it} + g(t) + u_i + e_{it},$$

and reports $\hat{\beta}_D$ as 'the treatment effect'. Taking the on-drug minus off-drug contrast that β_D targets, and using that the natural-history component is by construction independent of treatment status so that its on/off difference cancels in a within-participant comparison, the probability limit of the estimator is

$$\operatorname{plim} \hat{\beta}_D \;=\; m_{BR} + b_{PB} + b_{TV},$$

where $b_{PB} = E[PB \mid \text{on}] - E[PB \mid \text{off}]$ is the belief differential between the contrasted phases and b_{TV} collects any natural-history contribution that fails to cancel under the contrast actually used; b_{TV} is zero for a clean on/off within-participant contrast and strictly positive for a baseline-anchored pre-post analysis, in which case $b_{PB} = E[PB]$ and $b_{TV} = E[TV]$. To fix which case is in play below: the one-component analysis used in the worked example and the simulations of this paper is the design-contrast (drug-on/off) form, so $b_{TV} \approx 0$ there and the operative displacement is the belief term b_{PB}; the baseline-anchored pre-post form, which additionally carries $b_{TV} = E[TV]$, is the worse case retained only for comparison. Both terms are generally non-negative for an improving outcome, so the one-component estimate over-states the pharmacological component m_{BR}. The displacement is a property of the probability limit and is unchanged as $N \to \infty$: a larger one-component trial converges on the same biased value. This is the precise sense in which the placebo-attribution failure of section 1 is a problem of identification, not of precision.

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The same algebra settles the biomarker-validation failure. The precision-medicine analysis regresses a participant-level response summary on a baseline biomarker B_i. If that summary is the one-component per-participant effect, which estimates $m_{BR,i} + w_{PB}\, m_{PB,i} + w_{TV}\, m_{TV,i}$ for design-determined loadings $w_{PB}, w_{TV} \geq 0$, then by the linearity of covariance the fitted biomarker slope has probability limit

$$\beta_{bm}^{\text{lumped}} \;=\; \frac{\operatorname{Cov}(m_{BR} + w_{PB} m_{PB} + w_{TV} m_{TV},\, B)}{\operatorname{Var}(B)} \;=\; \beta_{bm}^{BR} + w_{PB}\, \beta_{bm}^{PB} + w_{TV}\, \beta_{bm}^{TV},$$

where $\beta_{bm}^{C} = \operatorname{Cov}(m_C, B)/\operatorname{Var}(B)$ is the component-specific slope. A biomarker correlated with placebo responsiveness or with the natural-history trajectory, so that β_{bm}^{PB} or β_{bm}^{TV} is non-zero, therefore registers a non-zero one-component slope even when its pharmacological association β_{bm}^{BR} is exactly zero. This is the biomarker-validation failure of section 1, and it too is an identity under the additive decomposition rather than an empirical tendency: no sample size separates the one-component slope from a genuine pharmacological one, because the two coincide in the estimand whenever $w_{PB}\beta_{bm}^{PB} + w_{TV}\beta_{bm}^{TV} \neq 0$.

We make two qualifications explicit, because they bound how much weight the framework can carry. First, both displays are immediate corollaries of the additive decomposition and the linearity of the projection that defines the estimator; they are not a derived result in any deeper sense, and their value is in naming the displacement terms, not in the algebra. Second, they assume the components combine additively. Under the subadditive interaction documented in section 3.3, where the joint effect of BR and PB falls below their sum, a first-order correction proportional to $\operatorname{Cov}(m_{BR} \cdot m_{PB}, B)$ enters the biomarker slope, and the identities above should be read as the leading behavior in the additive case rather than as exact under interaction. The empirical magnitude of that correction term is itself an open question that the framework's γ-interaction simulation (section 3.3) is designed to address.

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Two consequences for how the framework should be evaluated follow. First, the quantity that carries the bias is the location of the estimand, $\operatorname{plim} \widehat{\beta}_D$, not its sampling variability: the one-component coefficient remains a well-estimated number with a valid Wald test of its own nullity, and that test rejects at the rate dictated by the displaced value. A significance test of the one-component coefficient therefore cannot, in principle, detect the bias, because the bias does not move the coefficient toward zero. Detecting it requires comparing $\widehat{\beta}_D$ against m_{BR}, which is exactly the comparison the decomposition, or a phase contrast that isolates

m_{BR}, makes available and the one-component analysis does not. Second, and as a corollary for the simulation evidence, a rejection-rate or power summary is the wrong instrument for this bias: the right target is the coefficient itself and its offset from m_{BR}. This is why the pilot of section 7.1 found the biomarker-treatment rejection rate essentially flat across the placebo-strength axis even though that axis is precisely what drives the predicted bias; the prediction lives in the mean of $\widehat{\beta}_{bm:D_{bc}}$, and a production run that means to test it must read the bias off the coefficient, not off the test.

6.2 A numerical illustration

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[Figure 1 here: two-panel trajectory plot. Left panel shows open-label phase (weeks 1–8) with overlapping response trajectories for Participants A and B. Right panel shows blinded-discontinuation phase (weeks 9–12) with the diagnostic divergence: A retains four points while B collapses to 1.5. Component contributions are labeled in each panel.]

Two participants in a sixteen-week N-of-1 trial of prazosin for PTSD share the same baseline but differ in latent traits: Participant A has $m^A_{BR} = 4$, $m^A_{PB} = 6$; Participant B has $m^B_{BR} = 8$, $m^B_{PB} = 1$. Both reach approximately ten points of open-label improvement by week eight, making them indistinguishable to a one-component analysis that estimates a population drug effect of 7.75 points. The blinded-discontinuation phase supplies the identifying contrast: A retains four points of improvement (belief persists at $\eta = 0.5$) while B returns nearly to baseline (drug cleared, minimal belief). The three-component decomposition recovers all six component estimates by exploiting this phase differential; the one-component model cannot.

7 Simulation study

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The simulation study reported in this section is structured under the[33] ADEMP framework, specifying Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures. The working ADEMP protocol is at `analysis/scripts/component-decomposition/00-ademp-protocol.md`. The

study is exploratory: no analysis plan was registered in advance, and its findings should be confirmed in an independent study. A Monte Carlo simulation calibrated to the prazosin/PTSD reference parameters quantifies the bias and identifiability properties of the three analysis strategies described in section 5 across a parameter grid.

7.1 Pilot validation

[Details in supplement S1.]

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A 100-replicate pilot at $N = 35$ on a 6-cell grid confirmed the end-to-end pipeline. Two observations inform the production specification: at $N = 35$, rank-deficiency dropped the D_{bc}:phase interaction term, removing the key contrast; and the predicted bias appears in the coefficient mean, not the rejection rate, so the production program reports $\widehat{\beta}_{bm:D_{bc}}$ against its implied true value rather than power alone. Full details and exploratory summaries are in supplement S1.

7.2 Production-simulation specification

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The production program follows the[33] ADEMP framework and targets four deliverables: the bias of $\widehat{\beta}_{bm:D_{bc}}$ for the one-component, phase-augmented, and full-decomposition analyses across $N \in \{35, 70, 100, 150\}$ and a *pb_strength* grid; identifiability metrics for the full-decomposition fit (convergence, variance-inflation factors, and sensitivity of the BR estimate to the η(phase) multipliers); Type I error and power for the interaction test under an η-mismatch, TV-magnitude, and dropout sensitivity grid, with null cells at $n_{\text{reps}} = 5{,}000$ for MCSE 0.003; and the participant-level interaction signal recovered as a function of (m_{BR}, m_{PB}, m_{TV}), which the one-component-total regression does not preserve. The `tidyverse` implementation collection drives the grid; driver scripts, checkpointed cell-level summaries, per-replicate output, and the committed ADEMP protocol all reside under `analysis/scripts/component-decomposition/`. The realised Study A run reported next used 1,000 replicates per alternative cell.

7.3 Study A: orthogonal-biomarker arm (1,000 replicates per alternative cell)

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The production Study A run (1,000 replicates per alternative cell, 5,000 per null cell, 80 alternative cells crossing $m_{PB} \in \{0, 1, 3, 6, 10\}$, $m_{TV} \in \{-1, 0, 1, 2\}$, and $N \in \{35, 70, 100, 150\}$, plus four null cells) is the orthogonal-biomarker arm of the conditional evaluation. The data-generating process couples the baseline biomarker to the BR component only, through the on-drug biomarker-BR covariance channel, so the true biomarker-treatment effect is the biomarker-by-BR slope, $\beta_{bm:D_{bc}} = -c_{bm}\,\sigma_{BR}/\sigma_{bm} = -2.25$, and the biomarker is by construction uncorrelated with PB and TV. Under the one-component analysis the mean estimated $\widehat{\beta}_{bm:D_{bc}}$ tracks this value across the whole grid: the mean absolute bias is 0.016 at $N = 35$, 0.016 at $N = 70$, 0.006 at $N = 100$, and 0.006 at $N = 150$, in every case within one MCSE (respectively about 0.028, 0.019, 0.009, and 0.007) of zero, with no monotone trend across m_{PB} or m_{TV}. The predicted placebo-attribution bias of the biomarker-treatment estimate does not appear.

This is not a failure of the simulation; it is the behavior the identity of section 6.1 requires. That identity writes the one-component biomarker slope as $\beta_{bm}^{BR} + w_{PB}\beta_{bm}^{PB} + w_{TV}\beta_{bm}^{TV}$. With the biomarker coupled to BR alone, $\beta_{bm}^{PB} = \beta_{bm}^{TV} = 0$, so the one-component slope equals the pharmacological slope no matter how large m_{PB} or m_{TV} grows. Study A varies the magnitudes of the unmodeled components, which the identity shows to be the innocuous axis; the axis that produces biomarker-validation bias is the biomarker's correlation with PB or TV, which this DGP holds at zero. The clean null is therefore a confirmation of the identity and a correction to the looser claim of section 1 that large PB or TV by itself biases biomarker validation. It does not; biomarker-component coupling does.

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The comparison between analyses is equally clear. The phase-augmented analysis is not less biased than the one-component analysis at any sample size (its mean absolute bias is 0.028 versus 0.016 at $N = 35$ and is comparable elsewhere), and it is markedly less powerful for the biomarker-treatment test: mean power across the grid is 0.35 versus 0.59 at $N = 35$ and 0.62 versus 0.90 at $N = 70$, with both analyses reaching essentially full power by $N = 100$. Because the DGP contains no biomarker-PB contamination for the phase contrast to remove, the extra phase-by-treatment terms add estimation cost without buying any bias reduction, and

the analysis is dominated. Convergence was approximately complete for both analyses at every sample size (minimum convergence rate 0.998), so the rank-deficiency that forced dropping the `Dbc:phase` term in the $N = 35$ pilot, which we had advanced as the explanation for the pilot's inverted power ranking, is not operative in the production run: the inverted ranking persists with the full formula fitted and converged. The pilot's power inversion was therefore a genuine feature of this DGP, not an artifact of rank-deficiency.

Calibration is adequate for both analyses. Nominal-95% confidence-interval coverage of $\beta_{bm:D_{bc}}$ ranges from 0.96 to 0.99 for the one-component analysis (slightly conservative) and from 0.94 to 0.98 for the phase-augmented analysis. Type I error at the four null cells is controlled, ranging from 0.031 to 0.056 for the one-component analysis and 0.038 to 0.049 for the phase-augmented analysis; the one-component estimate at $N = 100$ (0.056, MCSE 0.003) sits 1.9 standard errors above nominal but is within the simulation's Monte Carlo uncertainty.

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Two conclusions follow for the Study A program. First, the result sharpens rather than supports the section 1 framing: the biomarker-validation failure is real as an identity, but it is driven by the biomarker's coupling to PB or TV, not by the magnitude of those components, and Study A as designed varies the latter while holding the former at zero. The decisive axis is the biomarker-PB correlation $c_{bm,PB}$, which the contaminated-biomarker pilot in the following subsection varies directly. Second, on the present evidence the phase-augmented analysis carries a real power cost without a compensating bias reduction under an uncontaminated biomarker, so it should not be recommended as a default for this design; the pilot below evaluates whether contamination changes this verdict.

7.4 Study B pilot: coupling-arm simulation (100 replicates per cell)

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A supplementary pilot varying the biomarker-PB correlation $c_{bm,PB} \in \{0, 0.15, 0.30, 0.45\}$ at two PB magnitude levels ($m_{PB} \in \{0, 6\}$) and two sample sizes ($N \in \{70, 150\}$), using 100 replicates per cell, confirms the identity-based prediction and reveals an unexpected structural result.

At $N = 150$ the one-component bias scales monotonically with $c_{bm,PB}$: mean $\hat{\beta}_{bm:D_{bc}}$ moves from -2.31 ($c_{bm,PB} = 0$, bias -0.06) to -2.37 ($c_{bm,PB} = 0.15$,

bias -0.12) to -2.08 ($c_{bm,PB} = 0.30$, bias $+0.17$) to -1.74 ($c_{bm,PB} = 0.45$, bias $+0.51$). The last value sits 30 MCSEs from the true -2.25, establishing the effect unambiguously. The direction of bias is positive (toward zero): high-$c_{bm,PB}$ participants experience stronger PB, which in the open-label phase adds to the apparent drug effect, but this elevation is partially absorbed by the bm main effect, leaving the $bm : D_{bc}$ interaction coefficient attenuated relative to the pure pharmacological value.

The unexpected result is that m_{PB} does not modulate the bias: cells with $m_{PB} = 0$ and $m_{PB} = 6$ produce identical bias at every contamination level (differences ≤ 0.006 at $N = 150$, well within Monte Carlo error). The driver is $c_{bm,PB} \times \sigma_{PB}$, not $c_{bm,PB} \times m_{PB}$. Even when the population-mean PB amplitude is zero, individual participants have nonzero PB draws (standard deviation $\sigma_{PB} = 2$), and coupling the biomarker to that variance is sufficient to contaminate the estimator. A true no-PB control requires $\sigma_{PB} = 0$, not $m_{PB} = 0$. This refines the Study A design: varying m_{PB} at $c_{bm,PB} = 0$ correctly produced zero bias, but because the relevant null was $c_{bm,PB} = 0$, not $m_{PB} = 0$.

The phase-augmented analysis provides no protection: its bias at every $(c_{bm,PB}, m_{PB}, N)$ cell is indistinguishable from the one-component bias. Adding phase-by-treatment interactions to the model does not decompose the contamination because both analyses see the biomarker-PB covariance through the same open-label versus blinded-discontinuation contrast; the contrast changes the weight on PB (via expectancy) but does not isolate the PB component's contribution to the biomarker slope. Separating that contribution requires both a design that decouples drug exposure from belief and a decomposition that exploits it; Study C shows that the standard hybrid design cannot achieve this separation, whereas a balanced-placebo design can.

The $N = 70$ results are uninformative at 100 replicates (Monte Carlo standard error ≈ 0.06, comparable to or larger than the bias at moderate $c_{bm,PB}$). The production run will use 1,000 replicates per cell to establish the $N = 70$ pattern and 5,000 null replicates to assess Type I error under contamination. Until the production run is complete, the quantitative bias estimates from this pilot should be treated as directionally informative but imprecise.

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The pilot results reframe the paper's practical guidance. The risk of biomarker-validation failure in an aggregated N-of-1 study arises not from the presence of a strong placebo response in the population but from the biomarker's correlation with individual-level placebo responsiveness. In a population where the baseline biomarker is uncorrelated with each participant's placebo-belief response

amplitude, the one-component estimator recovers the pharmacological interaction without bias, and the full decomposition machinery is not required. In a population where the biomarker covaries with psychological characteristics that predict placebo response – optimizm, prior treatment experience, suggestibility, expectancy at enrollment – the one-component analysis conflates pharmacological and placebo-belief prediction, and the scale of that conflation grows with the coupling strength. At $c_{bm,PB} = 0.45$, a value that equals the pharmacological coupling used throughout this paper, the one-component bias is approximately 0.51 points on the interaction scale, representing 23% of the true pharmacological effect. This is a practically meaningful distortion for any precision-medicine stratification decision.

The production run of the contaminated-biomarker study, at 1,000 alternative-cell replicates and 5,000 null-cell replicates, is required to confirm these estimates, establish the $N = 70$ pattern, and characterize Type I error under contamination. Results are pending.

7.5 Study C: recovery under a belief-decoupling design

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Study C asks whether the contamination demonstrated in the coupling-arm study can be removed by decomposition, and finds that the answer depends on the trial design. In the MVN process the biomarker-BR coupling acts only on drug and so loads on the drug-state contrast D_{bc}, whereas the biomarker-PB coupling scales with the expectancy η; interacting the biomarker with both isolates β_{bm}^{BR} on $bm{:}D_{bc}$, but only where D_{bc} and η vary independently. On the standard hybrid design they do not (on-drug time and belief accumulate together, with an analysis-stage correlation of about +0.6), and we verified that the belief-covariate decomposition, a blinded-stratum drug contrast, and a Gompertz-basis component decomposition all fail to recover β_{bm}^{BR} there. The remedy is a design that breaks the collinearity: a balanced-placebo (open-hidden) design[25,26] interposes a covert on-drug phase, in which the drug is given without the participant's knowledge ($\eta = 0$), and an open-placebo phase, in which the participant is off drug but believes treatment continues ($\eta = 1$), reducing the D_{bc}-η correlation to about -0.66.

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The production run (1{,}000 replicates per cell, $N \in \{70, 150\}$, $c_{bm,PB} \in \{0, 0.1, 0.2, 0.3\}$) confirms the recovery. At $N = 150$ the decomposition tracks the pharmacological target -2.25 across the contamination range, with bias -0.022, $+0.008$, $+0.021$, and -0.026, every value within one MCSE (about 0.02) of zero, while the one-component estimator develops a monotone bias of $+0.16$, $+0.34$, and $+0.48$ as $c_{bm,PB}$ grows. Interval coverage holds near nominal for the decomposition (0.94 to 0.96) and degrades to 0.86 for the one-component analysis. The power comparison (Table 2) is more nuanced than a uniform decomposition penalty: at zero contamination the one-component analysis is more powerful, by 0.19 at $N = 70$ (0.85 versus 0.66) and 0.07 at $N = 150$, reflecting the degrees of freedom the decomposition spends, but this advantage erodes monotonically as $c_{bm,PB}$ grows and reverses by $c_{bm,PB} = 0.3$ (0.64 versus 0.67 at $N = 70$). The reason is that contamination in this direction drags the one-component point estimate toward the null and so costs it power, whereas the decomposition holds the estimate at the pharmacological target; the one-component power advantage is accordingly confined to the low-contamination regime in which decomposition is unnecessary. Recovery is therefore conditional on the design, not on the analysis alone: the decomposition is necessary but a design that varies belief independently of drug exposure is the enabling condition. The practical recommendation is that a biomarker-validation trial in which the candidate biomarker may correlate with placebo responsiveness should be fielded as a balanced-placebo or open-hidden design rather than the standard hybrid design. Per-replicate output is at `analysis/data/derived_data/component-decomposition/study-b-balanced/`. Recovery beyond the positive-definite range, participant-level component recovery, and replication under the mean-moderation architecture coupling remain open.

Table 2: Rejection rate (power) for the biomarker-treatment interaction at $\alpha = 0.05$ under the balanced-placebo design, one-component versus belief-covariate decomposition, by contamination $c_{bm,PB}$ and sample size (1,000 replicates per cell; paired standard error of the gap ≤ 0.014). The one-component advantage at $c_{bm,PB} = 0$ erodes and reverses as contamination grows, because contamination drags the biased one-component estimate toward the null.

$c_{bm,PB}$	N	One-component	Decomposition	Gap
0.0	70	0.85	0.66	+0.19
0.1	70	0.78	0.67	+0.11
0.2	70	0.74	0.69	+0.05
0.3	70	0.64	0.67	−0.03
0.0	150	0.99	0.93	+0.07
0.1	150	0.98	0.93	+0.05
0.2	150	0.97	0.93	+0.04
0.3	150	0.93	0.93	−0.00

8 Application to predictive-biomarker validation

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The motivating clinical question for the trial designs in this project is whether a baseline biomarker predicts treatment response, and specifically whether the biomarker predicts the pharmacological component of treatment response. This is the precision-medicine question, formalized in the PATH statement[8]. Whether it depends on the BR-PB-TV decomposition is itself conditional: by the identity of section 6.1, the one-component biomarker slope equals the pharmacological slope when the biomarker is uncorrelated with PB and TV, and departs from it only by the biomarker's covariance with those components. The decomposition is therefore the mechanism the question requires precisely in the regime where that covariance is non-negligible.

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A direct regression of total response on baseline biomarker

```
total_response = beta_0 + beta_bm * biomarker + error
```

estimates the biomarker's correlation with whatever the one-component total contains. If PB varies systematically across the biomarker distribution, because the biomarker is correlated with optimizm, suggestibility, or treatment-seeking history, then $\widehat{\beta}_{bm}$ contains a placebo-prediction component inseparable from the pharmacological-prediction component. The resulting biomarker, applied prospectively, would mis-stratify patients: those flagged as 'high responders' would include both true high-BR responders and high-PB responders, and the latter would not benefit from prescription decisions made on a pharmacological basis.

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The right approach is to regress each component on the biomarker:

$$
\begin{aligned}
BR &= \beta_0^{BR} + \beta_{bm}^{BR} \cdot \text{biomarker} + \varepsilon^{BR}, \\
PB &= \beta_0^{PB} + \beta_{bm}^{PB} \cdot \text{biomarker} + \varepsilon^{PB}, \\
TV &= \beta_0^{TV} + \beta_{bm}^{TV} \cdot \text{biomarker} + \varepsilon^{TV}.
\end{aligned}
$$

A biomarker is a useful pharmacological predictor if and only if $\hat{\beta}_{bm}^{BR}$ is meaningfully non-zero. Significant $\hat{\beta}_{bm}^{PB}$ or $\hat{\beta}_{bm}^{TV}$ are diagnostic findings about the biomarker, indicating that it is partly psychological or correlates with natural-history characteristics, but they are not the validation of pharmacological prediction. A biomarker that correlates only with PB would be valuable for selecting placebo responders in run-in designs but worthless for predicting pharmacological efficacy; a biomarker that correlates only with TV would be a prognostic marker, not a predictive one.

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The decomposition is therefore the mechanism by which the precision-medicine question is posed whenever the candidate biomarker may correlate with PB or TV. Where the biomarker can be assumed orthogonal to the non-pharmacological components, the identity of section 6.1 shows the one-component analysis already targets the biomarker-by-BR estimand, and the simulation of section 7.3 confirms it recovers that estimand without bias. The phase-augmented linear-mixed analysis introduced in section 5 retains a useful subset of the decomposition machinery at small N (the `bm:Dbc:phase` interaction tests whether the biomarker-treatment effect attenuates under blinding, the practical proxy for biomarker-by-PB contamination), but Study A shows this proxy earns its degrees of freedom only when such contamination is present: under an uncontaminated biomarker it reduced no bias and lost power. The full decomposition recovers the component-specific regressions explicitly when N permits and when the scientific question warrants the cost.

9 Discussion

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The three-component decomposition is the formal expression of the claim that 'treatment response' in an aggregated N-of-1 trial is not a single quantity but a sum of three causally distinct contributions, each with its own pharmacological, psychological, or epidemiological interpretation. What the present work establishes, analytically in section 6.1 and empirically in section 7.3, is that the inferential value of acting on this decomposition is conditional rather than universal. The identity shows that the one-component estimator is displaced from its pharmacological target only by the components' loading onto the contrast used, and, for

a biomarker slope, only by the biomarker's covariance with PB and TV. Where those terms are zero, the design-contrast one-component analysis is already unbiased, and the production simulation confirms it: with a biomarker coupled to BR alone, the one-component biomarker-treatment estimate is unbiased across the whole PB-TV grid and the phase-augmented analysis is dominated. The framework's contribution is therefore best read as a map of when decomposition changes inference, not as a blanket prescription to decompose. It makes the trial-design conditions explicit, identifies the attribution and contaminated-biomarker estimands for which decomposition is needed, and clarifies the target estimand for biomarker validation (the biomarker-by-BR association, not the one-component total).

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Four open questions remain and warrant acknowledgement. First, the simulation study described in section 7 is forthcoming; the present manuscript reports the framework rather than its empirical evaluation, and the quantitative bias-versus-power trade-offs across analysis strategies are conjectures until that work is run. Second, the η(phase) multipliers used to parameterize the expectancy modulation of PB are themselves modeling assumptions, and a sensitivity analysis varying η across plausible values is mandatory before any clinical conclusion is drawn from the framework's outputs. Third, the framework as specified treats the three components as monotone Gompertz trajectories; cyclical or biphasic TV patterns require either flexible TV specifications or explicit covariates beyond the Gompertz form, and the boundary between 'flexible-Gompertz' and 'covariate-augmented' specifications is not yet sharply drawn. Fourth, the participant-level component estimates the framework targets inherit the identifiability caveat that[29] and[28] raise for individual treatment effects more generally: within-participant component recovery rests on separating a true participant-by-component signal from participant-by-period and participant-by-phase variation, and this separation is credible only when the design supplies repeated phase contrasts and the cross-component covariance structure is not pathological. The framework should be read as estimating these quantities under those assumptions, not as dissolving the well-known difficulty of estimating individual effects from finite within-person data.

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The companion pedagogical document at `docs/24-component-decomposition-pedagogy.md` develops the framework with worked examples and an extended intuition narrative for readers approaching the material for the first time. The companion methodological evaluation of the Gompertz family at `analysis/report/07-gompertz-evaluation/` quantifies the sensitivity of the framework's inference to the specific parametric form chosen for the component trajectories.

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A final point concerns the scope of the decomposition, where it is essential to distinguish the existence of the components from their identifiability. The claim that an observed longitudinal treatment response is the sum of a pharmacological, a belief-driven, and a natural-history component is a statement about the data-generating process of any repeated-measures treatment study, and in that sense the decomposition is not specific to N-of-1 trials. Identifiability, by contrast, is design-specific, and the distinction between population-level and participant-level recovery is decisive for the biomarker application. A standard parallel-group placebo-controlled trial supplies a between-arm placebo contrast that identifies the population-mean belief response but not a participant-level PB, and it contains no within-subject contrast in which belief varies independently of pharmacology; the participant-level biomarker-by-BR slope that the precision-medicine question targets is therefore not identified from such a design, no matter how large it is. The designs that do supply the required within-subject belief contrast are the blinded or open-hidden discontinuation and randomized-withdrawal designs[25,26], the crossover with on/off periods, and the hybrid N-of-1 design developed here; the pharmacometric disease-progression-plus-placebo models from which the framework descends[1,2,34] operate at the group level and recover population-mean components only. We therefore advocate the decomposition as a default lens for longitudinal designs that supply a within-subject belief-by-exposure contrast, with the aggregated N-of-1 case the most demanding instance because it must recover the components from a single participant's trajectory. We do not claim participant-level identifiability from parallel-group placebo control alone, and where only a between-arm placebo contrast is available the framework's claims are restricted to the population mean.

10 Conclusions

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In conclusion, three points are to be emphasized. First, treatment response in aggregated N-of-1 clinical trials is the sum of three causally distinct components: pharmacological action (BR), placebo-belief response (PB), and time-variant natural history (TV). Each component has its own biological mechanism, its own identifiability conditions, and its own clinical interpretation. Whether conflating them biases a given analysis is, however, governed by the omitted-variable-bias identity of section 6.1, not by the raw magnitude of PB and TV: the one-component pre-post treatment effect is displaced by the PB and TV contributions, whereas the biomarker-treatment slope is displaced only by the biomarker's covariance with those components. Study A bears this out: varying PB and TV magnitude with a biomarker coupled to BR alone left the one-component estimate unbiased across the entire grid.

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Second, the operative requirement is the design contrast, not the parametric analysis model. The trial design must include the drug-on/off and belief contrasts that identify the components, and the one-component analysis already exploits the drug-on/off contrast through its D_{bc} term. Adding parametric component structure on top is not free: under an uncontaminated biomarker the phase-augmented analysis reduced no bias and lost substantial power, so it should not be adopted as a default. Its value, and that of the full component-matched fit, is confined to the estimands where the extra structure does work, namely the attribution of response to pharmacology versus belief and the analysis of biomarkers that may themselves correlate with PB or TV.

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Third, for predictive-biomarker validation the decomposition earns its place precisely when the candidate biomarker may correlate with PB or TV. A biomarker that predicts the one-component total is a pharmacological predictor only when it is uncorrelated with the non-pharmacological components; where that cannot be assumed, the one-component slope mixes pharmacological and non-pharmacological prediction, and the regulatory and prescribing decisions that motivate the trial require the separation the decomposition supplies. The coupling-arm study demonstrates this failure directly, and Study C establishes that the bias is recoverable: a belief-covariate decomposition returns the unbiased pharmacological slope across the feasible contamination range, but only on a balanced-placebo design that decouples drug exposure from belief, not on the standard hybrid design. The decisive practical lesson is therefore as much about design as about analysis.

11 Conflict of interest

The authors declare no conflicts of interest.

12 Funding

This work received no specific funding.

13 Data availability statement

The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

14 Reproducibility

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This manuscript was rendered with R 4.4.x inside the project's zzcollab Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `parallel` (8-core `mclapply` for parameter-grid sweeps), `data.table` (the `original-extended` simulation collection), `furrr` and `purrr` (the `tidyverse` simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build.

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Each simulation driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`. Per-replicate seeds derive from the master seed by `+ rep_idx`.

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Per-replicate simulation outputs are checkpointed at `analysis/data/quick-sim/06-decomp-replicates`. Morris ADEMP cell-level summaries at the companion `06-decomp-summary.txt`. Driver scripts are under `analysis/scripts/component-decomposition/`. The working ADEMP protocol is at `analysis/scripts/component-decomposition/00-ademp-protocol.md`. The manuscript source, paper-specific bibliography (`references.bib`), and SIM CSL file are at `analysis/report/06-component-decomposition/`.

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